

Response to Reviewers

Manuscript Jc26-0002: “Semantic-Topological Cognitive Maps for Assistive Indoor Navigation of a Robot Partner”

Anhar Risnumawan, Achmad Fahrul Aji, Raditya Artha Rochmanto,
Teuku Zikri Fatahillah, and Naoyuki Kubota

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Dear Editor-in-Chief, Associate Editor, and Reviewers,

We thank the Associate Editor and both reviewers for the careful reading and constructive feedback on our manuscript. The comments have materially improved the rigor, clarity, and presentation of the work. Below we address every comment point-by-point. Comments are reproduced verbatim in italics; our responses and the corresponding manuscript changes follow each comment, with references to sections, tables, and figures in the revised manuscript. All revisions in the manuscript are marked with track changes.

Contents

I. Associate Editor: Summary Requirements

AE-1. Multi-scenario quantitative evaluation

***Comment:** Supplement critical quantitative evaluation metrics, including but not limited to navigation success rate, semantic labeling accuracy (e.g., mIoU, F1-score), and natural language grounding accuracy. All data must be derived from at least three independent experiments conducted across multiple indoor scenarios (e.g., offices, corridors, meeting rooms), with detailed descriptions of the dataset provided.*

Response: We thank the Associate Editor for emphasizing the need for stronger quantitative validation. The evaluation has been expanded to **three independent experimental sessions** in three distinct indoor scenarios: (i) a corridor, (ii) an office/lab room, and (iii) a meeting room. For each scenario we report a shared protocol covering: (a) navigation success rate with 95% confidence intervals, completion time, and path-length ratio; (b) semantic labeling accuracy in terms of macro-F1 and micro-F1 on stable node labels under a fixed evaluation vocabulary; and (c) natural-language grounding accuracy on a set of ≈ 15 commands per scenario. The dataset and annotation protocol are described in the new **Sec. 5.1 (Dataset and Evaluation Protocol)**, and the headline metrics are summarized in the new **Table A (dataset summary)** and **Table B (main quantitative results across three scenarios)** in Sec. 5.2.

AE-2. SOTA comparison

***Comment:** Conduct comparative experiments with 3–5 representative state-of-the-art (SOTA) methods (e.g., SemanticFusion, HG-Nav, OV-Map). Provide both quantitative results and qual-*

itative analyses (e.g., map completeness, environmental adaptability) to highlight the advantages and differences of your proposed STCM framework.

Response: We fully agree on the importance of broader comparison. Because the cited representative systems (SemanticFusion, OV-Map/OneMap, LM-Nav/NLMap, Kimera/S-Graphs+, HG-Nav/NTS/RoboHop) rely on heterogeneous sensing (dense RGB-D vs. sparse LiDAR), map representations (dense volumetric vs. sparse place graph), and execution backends (custom planners vs. ROS 2/Nav2), a forced full end-to-end reproduction on our platform would not yield a scientifically fair comparison within this revision cycle. We therefore adopted a **hybrid comparison strategy**:

- **Direct quantitative comparison** against three internal executable baselines evaluated on the same data under the same protocol: *semantic-only* (no topological abstraction), *place-GNG-only* (topology without instance-GNG), and *no-LLM* (label matching with simple templates). Results are in the new **Table B** and **Table C**.
- **Capability-oriented SOTA table** (new **Table F**) that positions STCM against SemanticFusion, OV-MAP/OneMap, Kimera/S-Graphs+, LM-Nav/NLMap, and NTS/RoboHop along six axes: sensor assumption, map representation, online operation, open-vocabulary queryability, language-to-graph grounding, and deployment backend.
- **Qualitative analysis** of map completeness, environmental adaptability, and interpretability is provided in Sec. 5.5.

This hybrid design is described transparently, and we explicitly state that we avoid unfair end-to-end reproduction claims (see Sec. 5.1 and the Discussion).

AE-3. LLM ablation with statistical analysis

Comment: Design ablation experiments to explicitly demonstrate the performance improvement brought by the LLM component. Statistical analysis (e.g., *t*-tests) should be used to verify the significance of the improvements compared to a baseline system without the LLM module.

Response: We clarified that the LLM in STCM performs *language-to-graph grounding only* and is *not* an end-to-end navigation policy (revised Sec. 4). A dedicated LLM ablation has been added as new **Sec. 5.4 (LLM Ablation)** and **Table C**. STCM+LLM is compared with a no-LLM baseline (label string match + simple templates) on the same command set (≈ 45 commands total, split into *simple label*, *disambiguation*, and *spatial/compositional* subsets). We report overall exact-match accuracy and relational-subset accuracy, together with paired per-scenario statistics: effect size and **McNemar’s test** *p*-value per command outcome, with 95% confidence intervals. The LLM’s temperature is fixed at 0 and prompts plus graph JSONs are archived for reproducibility.

AE-4. Computational performance and real-time feasibility

Comment: Disclose the system’s computational performance parameters, including latency of key modules, frame rate (FPS), and hardware configuration (CPU/GPU model, memory size). Provide a detailed analysis to confirm the feasibility of real-time operation for assistive navigation.

Response: We added **Table D (runtime and hardware disclosure)** in the new **Sec. 5.6 (Computational Performance)**. It reports, per module (FAST-LIO2, dual-GNG place learning, dual-GNG instance learning, GroundingDINO+SAM, LiDAR–image association, temporal fusion, LLM grounding, Nav2 planning/control): mean and p95 latency over ≥ 100 iterations, effective rate (Hz/FPS), and CPU/GPU/RAM usage. The hardware configuration (CPU model, GPU model, memory size, ROS 2 distribution) is listed explicitly. The analysis separates the

navigation-critical loop (FAST-LIO2, Nav2) which runs in real time, from *asynchronous modules* (semantic perception, LLM grounding) that update at lower rates without blocking navigation.

AE-5. Abbreviations and glossary

Comment: Provide the full form of all abbreviations (e.g., SLAM, LLM, LIFGIF, STCM, GNG) when they first appear in the manuscript (including the Abstract and Introduction). A glossary of abbreviations is recommended to improve readability.

Response: All major abbreviations are now defined at first use in both the Abstract and the Introduction, including SLAM (simultaneous localization and mapping), LLM (large language model), STCM (Semantic-Topological Cognitive Map), GNG (Growing Neural Gas), LiDAR (Light Detection and Ranging), LIO (LiDAR-inertial odometry), ROS 2 (Robot Operating System 2), Nav2 (Navigation2), SAM (Segment Anything Model), FAST-LIO2 (Fast Direct LiDAR-Inertial Odometry 2), and FPS (frames per second). The ambiguous acronym “LIFGIF” has been **removed** and replaced with its descriptive phrase throughout, since a canonical expansion was not available in the cited source. A consolidated **glossary/abbreviation table** has been added as Table G in the front matter.

AE-6. Figure resolution and text size

Comment: Adjust the text size in all figures to a minimum of 10pt and submit high-resolution images (resolution of no less than 300 dpi) to ensure clarity of labels, legends, and annotations (especially for Fig. 9).

Response: All figures have been regenerated at ≥ 300 dpi with a minimum in-figure font size of 10 pt. **Fig. 9 has been redesigned and split into two subfigures:** Fig. 9(a) shows the global STCM overlay on the metric map, and Fig. 9(b) shows the isolated place/instance graph view with enlarged labels and legend. Raster overlays were rebuilt from vector sources rather than upscaled.

AE-7. Shorten Introduction

Comment: Shorten the “Introduction” (to 1.5–2 double-spaced pages). Remove redundant background descriptions and focus on core issues, limitations of existing methods, and the innovations of your work.

Response: The Introduction has been shortened to approximately 1.8 double-spaced pages. Repeated cognitive-science exposition and paper-by-paper literature discussion have been removed. The revised Introduction retains only: (i) the assistive-navigation problem and the gap between metric SLAM and human goal specification; (ii) one compact paragraph identifying what prior work does not provide; and (iii) one concise contributions paragraph aligned with the refined four-point contribution list (see AE-10).

AE-8. Shorten Related Work

Comment: Shorten the “Related Work” (to 2–2.5 double-spaced pages). Remove redundant background descriptions and focus on core issues, limitations of existing methods, and the innovations of your work.

Response: Related Work has been condensed to approximately 2.3 double-spaced pages and restructured into four thematic paragraphs: (1) semantic / scene-graph mapping, (2) open-vocabulary mapping, (3) language-grounded navigation, and (4) topological navigation. Long

paper-by-paper summaries were replaced with a short closing paragraph that states precisely what prior work still does not provide and how STCM addresses it.

AE-9. Table 1 parameter rationale and sensitivity

Comment: *Explain the rationale for the parameter selection in Table 1, including references to relevant literature, parameter optimization processes (e.g., grid search), and sensitivity analysis results to demonstrate the robustness of the chosen parameters.*

Response: Table 1 has been expanded with a **rationale column** citing design constraints or relevant literature per parameter, and a compact **sensitivity study** (new **Table E** and accompanying plot) has been added for the four highest-impact parameters: D_{new} (place-node creation radius; sweep 0.3 / 0.5 / 0.8 m), α (temporal fusion gain; sweep 0.05 / 0.1 / 0.2), $\tau_{\text{box}} = \tau_{\text{text}}$ (detection thresholds; sweep 0.45 / 0.55 / 0.65), and η_{min} (minimum observations per instance; sweep 1 / 3 / 5). Results show a stable operating region around the selected values. GNG adaptation parameters (ϵ , age) follow standard GNG-style ratios and are justified by literature. We have **explicitly clarified that $D_{\text{gate}} = 0$ corresponds to the gating being disabled** in the reported runs; this is now stated in both Table 1 and the revised Method section to avoid ambiguity.

AE-10. Refine contributions

Comment: *Refine the research contributions into 3–4 specific and novel points, clearly distinguishing between technical innovations and engineering implementations. Avoid vague statements and emphasize the practical value of your work.*

Response: The contribution statement has been rewritten as four specific points that separate technical innovations from system-integration contributions:

1. **Online semantic-topological place graph:** an online STCM that turns a LiDAR–inertial SLAM pose stream into revisitable place nodes with explicit semantic-score vectors and graph connectivity for interpretable place-based goal selection and planning.
2. **Dual-GNG online learning:** a dual Growing Neural Gas formulation that separates place-topology learning from persistent object-instance consolidation, reducing duplicates and stabilizing semantic evidence under revisits.
3. **Map-frame open-vocabulary grounding:** anchoring GroundingDINO + SAM detections to the SLAM map frame through LiDAR–image association and temporal fusion, producing explicit node- and object-level semantic grounding.
4. **Deployable language-to-graph interface:** a ROS 2/Nav2-compatible interface in which the LLM grounds user commands to explicit STCM nodes instead of generating low-level control, enabling inspectable assistive navigation behavior.

AE-11 to AE-13. Submission package

Comment: *Attach a detailed “Response to Reviewers” document with page/section/table mappings; mark revisions with track changes; ensure formatting compliance.*

Response: The present document is the point-by-point response with explicit section, table, and figure references throughout. The revised manuscript is submitted with track changes enabled, and a final formatting pass was performed against the Fujipress/JACIII template (font size, line spacing, reference style, and figure captions).

II. Reviewer 1

R1-1. Quantitative evaluation metrics

***Comment:** The paper lacks quantitative evaluation metrics, such as navigation success rate, semantic labeling accuracy, or other performance indicators that could objectively assess the effectiveness of the proposed method.*

Response: We thank the Reviewer for this key observation. The revised manuscript now reports **navigation success rate** (with 95% CI), **semantic labeling accuracy** (macro/micro-F1 with label coverage), and **natural-language grounding accuracy** (exact-match and top-2) across **three indoor scenarios** under a shared evaluation protocol (≥ 30 navigation trials and ≈ 45 commands in total). Please see Sec. 5.1 for the protocol, Sec. 5.2 and Table B for navigation and semantic metrics, and Sec. 5.3 and Table C for grounding results. See also the response to AE-1.

R1-2. Experimental comparison with existing systems

***Comment:** Although the paper discusses related work extensively, it does not provide experimental comparisons between STCM and existing semantic mapping or topological navigation systems.*

Response: We added: (i) **direct quantitative comparison** against three internal executable baselines on the same data (semantic-only, place-GNG-only, no-LLM) in Table B and Table C; and (ii) a **capability-oriented SOTA table** (Table F) positioning STCM against Semantic-Fusion, OV-MAP/OneMap, Kimera/S-Graphs+, LM-Nav/NLMap, and NTS/RoboHop. We explain in Sec. 5.1 and the Discussion why a forced end-to-end reproduction of these heterogeneous systems would not yield a like-for-like comparison, and we therefore combine direct baselines with a transparent capability matrix. See also the response to AE-2.

R1-3. Extent of LLM contribution

***Comment:** The paper introduces an LLM-based reasoning interface; however, it remains unclear to what extent the LLM component improves the overall system performance.*

Response: We clarified the LLM’s role: it performs language-to-graph grounding only, not end-to-end control (revised Sec. 4). The new LLM ablation (Sec. 5.4, Table C) compares STCM+LLM vs. a no-LLM baseline on the identical command set, split into simple, disambiguation, and compositional subsets. The LLM yields its largest gain on ambiguous and compositional commands, while simple label-only commands already perform well without it. Paired McNemar statistics and effect size are reported. See also the response to AE-3.

R1-4. Computational performance

***Comment:** It would also be beneficial to report the computational performance of the system to clarify whether the proposed approach can support real-time operation.*

Response: The revised manuscript now includes the hardware configuration and a per-module runtime table (Table D, Sec. 5.6) with mean and p95 latency, effective Hz/FPS, and CPU/GPU/RAM usage. We explicitly separate the navigation-critical loop (FAST-LIO2, Nav2) which runs in real time, from the asynchronous semantic and LLM modules. See also the response to AE-4.

III. Reviewer 2

R2-1. Abbreviations at first use

Comment: *When an abbreviation is first used, the full form should be provided. For example, “SLAM”, “LLM” and “LIFGIF”.*

Response: All major abbreviations (SLAM, LLM, STCM, GNG, LiDAR, LIO, ROS 2, Nav2, SAM, FAST-LIO2, FPS) are now defined at first occurrence, including in the Abstract and Introduction. The acronym “LIFGIF” has been **removed** and replaced by its descriptive phrase, because a clean canonical expansion was not available in the cited source. A glossary table has been added. See also the response to AE-5.

R2-2. Figure text size, especially Fig. 9

Comment: *The text in figures is too small to read clearly, such as Fig. 9.*

Response: All figures have been regenerated at ≥ 300 dpi with a minimum in-figure font size of 10 pt. Fig. 9 has been redesigned from vector sources and split into two subfigures (global overlay and isolated graph view) to enlarge labels and legends. See also the response to AE-6.

R2-3. Introduction and Related Work are too long

Comment: *The lengths of “Introduction” and “Related Work” are too long. It is recommended to shorten them appropriately.*

Response: Introduction has been cut to ≈ 1.8 double-spaced pages and Related Work to ≈ 2.3 double-spaced pages. Redundant background discussion and paper-by-paper summaries were removed; the remaining text focuses on the problem, gap, and STCM’s specific contribution. See also the responses to AE-7 and AE-8.

R2-4. Basis for Table 1 parameter selection

Comment: *What is the basis for the selection of parameters in Table 1?*

Response: Table 1 has been expanded with an explicit rationale column (design constraint or literature basis) for every parameter, and a compact sensitivity study (Table E) has been added for the four highest-impact parameters: D_{new} , α , $\tau_{\text{box}}/\tau_{\text{text}}$, and η_{min} . The results show that the chosen settings lie within a stable operating region. The entry $D_{\text{gate}} = 0$ is now explicitly stated to mean that the gating was disabled in the reported runs. See also the response to AE-9.

R2-5. Contributions need refinement

Comment: *The contributions need to be further refined.*

Response: The contribution statement has been rewritten as four specific and more narrowly scoped points, separating technical innovations (online place graph, dual-GNG, map-frame open-vocabulary grounding) from the deployable ROS 2/Nav2 language-to-graph interface. See the revised Introduction and the list under the response to AE-10.

IV. Summary of new/changed assets

Asset	Where in revised manuscript	Addresses
Table A — Dataset / scenario summary	Sec. 5.1	AE-1, R1-1
Table B — Main quantitative results (3 scenarios)	Sec. 5.2	AE-1, AE-2, R1-1, R1-2
Table C — LLM ablation (with McNemar test)	Sec. 5.4	AE-3, R1-3
Table D — Runtime and hardware disclosure	Sec. 5.6	AE-4, R1-4
Table E — Parameter rationale and sensitivity	Sec. 4.x	AE-9, R2-4
Table F — SOTA capability matrix	Sec. 5.5	AE-2, R1-2
Table G — Abbreviation glossary	Front matter	AE-5, R2-1
Fig. 9 redesigned (split into 9a/9b)	Sec. 5	AE-6, R2-2
Shortened Introduction / Related Work	Sec. 1, Sec. 2	AE-7, AE-8, R2-3
Refined 4-point contributions	Sec. 1	AE-10, R2-5

We once again thank the Associate Editor and the reviewers for their constructive input, which substantially improved the rigor and clarity of this work. We hope the revised manuscript is now suitable for publication in JACIII.

Sincerely,
The Authors